

1. Administrative & Study Identification

Official Study Title: A Phase 2, Multicenter, Single-blinded, Randomized Study to Evaluate the Pharmacokinetics and Safety of Sotatercept (MK-7962) Administered Using Either a Weight-based or Weight-banded Approach in Participants With Pulmonary Arterial Hypertension (PAH) on Standard of Care **Public Study Title:** A Clinical Study of Sotatercept (MK-7962) in People With Pulmonary Arterial Hypertension (PAH) (MK-7962-024) **Short Title/Acronym:** LIGHTRAY (MK-7962-024) **Protocol Number:** MK-7962-024 **NCT Number:** NCT06664801 **Sponsor Name:** Merck **Investigational Product:** Sotatercept (MK-7962) **Phase of Development:** Phase 2 **Trial Status:** Active, Not Recruiting **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the pharmacokinetics (what happens to different doses of the drug in a person's body over time) and the safety/tolerability of sotatercept when administered using a new weight-banded dosing approach compared to the traditional weight-based approach. **Secondary Objectives:** To assess the initial dose pharmacokinetics and evaluate long-term tolerance when used alongside standard PAH background therapy.

2.2 Background & Rationale In Pulmonary Arterial Hypertension (PAH), blood vessels in the lungs become thick and narrow, restricting blood flow. Sotatercept is a targeted therapy working on proteins that cause PAH. Traditionally, the drug is dosed based on a patient's exact body weight. This study aims to determine whether grouping patients into "weight bands" (administering doses based on weight ranges) is a safe and pharmacokinetically equivalent alternative, which could significantly simplify clinical administration and optimize the management of PAH patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Weight-based doses vs. Weight-banded doses) **Blinding:** Single-blinded **Randomization:** Randomized (1:1 equal chance assignment)

3.2 Planned Enrollment & Duration Target **\$N\$:** 160 participants
Number of Sites: Multicenter (Global) **Estimated Study Start:** 14-Nov-2024 **Estimated Primary Completion:** 10-Jul-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Documented diagnostic Right Heart Catheterization with a World Health Organization (WHO) PAH Group 1 diagnosis in any of the following subtypes: Idiopathic PAH, Heritable PAH, Drug/toxin-induced PAH, PAH associated with connective tissue disease, or PAH associated with simple, congenital systemic-to-pulmonary shunts at least 1 year following repair. 3. Has symptomatic PAH classified as WHO Functional Class II or III. 4. Must have taken standard background PAH treatments for at least 3 months prior to starting the trial.

4.2 Exclusion Criteria 1. Body weight < 35 kg. 2. Diagnosis of PH WHO Groups 2, 3, 4, or 5. 3. Diagnosis of specific PAH Group 1 subtypes: HIV-associated PAH, PAH associated with portal hypertension, schistosomiasis-associated PAH, pulmonary veno-occlusive disease, and pulmonary capillary hemangiomatosis. 4. Cardiovascular conditions: Uncontrolled systemic hypertension; history of pneumonectomy; known pericardial constriction; restrictive cardiomyopathy; history of atrial septostomy within 180 days; personal or family history of long QT syndrome; history of coronary artery disease (myocardial infarction, percutaneous coronary intervention, coronary artery bypass, graft surgery, or cardiac anginal chest pain) within 6 months; cerebrovascular accident within 3 months; significant ($> 2+$ regurgitation) mitral regurgitation or aortic regurgitation valvular disease. 5. Untreated more than mild obstructive sleep apnea. 6. Known malignancy that is progressing or has required active treatment within the past 5 years. 7. Has recently started (within 12 months) or has plans to start weight loss medication or enter a weight loss program during the study period. 8. Has a previous (within 3 years) or planned (during the study) obesity treatment with surgery or a weight-loss device.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered once every 3 weeks. Both groups receive a lower first dose that slowly increases until the target dose is reached. The weight-banded group is put into 1 of 5 groups based on weight ranges, with the dose escalating up to a maintenance dose of 90 mg depending on the band. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Approximately 6 to 9 months (with an option to enter the MK-7962-031 extension study upon completion).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care background PAH treatments (medicines that help lower blood pressure in the lungs and improve blood flow). **Prohibited:** Other investigational drugs or specific excluded medications impacting cardiovascular hemodynamics outside of standard PAH therapy.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Right Heart Catheterization review, baseline weight measurement | | **Baseline** | Day 0 | Randomization, First SC Injection (lower starting dose), PK/Safety Labs | | **Interim** | Every 3 Weeks | SC Injections, Dose Titration (up-titration if tolerated), PK Sampling, AE Assessment | | **End of Study** | Week 24 - 36 | Final PK/Safety evaluation, transition to LIGHTRAY EXTENSION (MK-7962-031) if eligible |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Steady-State Average Serum Concentrations of Sotatercept (Cavg):** Steady-state Cavg of sotatercept will be reported. 2. **Number of Participants Who Experience One or More Adverse Events (AEs):** An AE is a health problem that happens or worsens during a study. Number of participants who experience an AE will be reported. 3. **Number of Participants who Discontinue Study Treatment due to an Adverse Event:** Number of participants who discontinue study treatment due to health problems that happen or worsen during the study will be reported.

7.2 Secondary & Exploratory Endpoints 1. **Initial Dose Average Serum Concentrations of Sotatercept (Cavg):** Evaluating the early pharmacokinetic processing of the drug. 2. **Incidence of dose reductions** required due to lack of tolerability.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for any untoward medical occurrence temporally associated with the drug. Particular attention given to common sotatercept AEs (such as epistaxis, telangiectasia, and dizziness). **Serious Adverse Events (SAEs):** Must be reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Regular safety monitoring to oversee dose up-titrations and verify safety between the dosing methodology arms.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: * *Safety Set*: All randomized participants who receive at least one dose of the study drug. * *PK Analysis Set*: Participants providing sufficient blood samples to evaluate pharmacokinetic comparability between the weight-based and weight-banded groups. **Sample Size Justification**: Target $N=160$, adequately powered to determine the pharmacokinetic equivalence and safety profile of the weight-banded dosing strategy. **Handling of Missing Data**: Standard multiple imputation methods for safety events; pharmacokinetic data processed per standard MK-7962 modeling algorithms.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan**: Routine onsite and remote monitoring to verify strict adherence to the assigned weight-based or weight-banded dosing calculations. **Audit Trail**: eCRF locking and stringent data validation for PK sample tracking and dose titration records.

11. Study Identifiers & Governance

Primary ID: MK-7962-024 **Registry Number**: NCT06664801 **Sponsor/Lead Organization**: Merck **Study Title**: LIGHTRAY **Official Regulatory Title**: A Phase 2, Multicenter, Single-blinded, Randomized Study to Evaluate the Pharmacokinetics and Safety of Sotatercept (MK-7962) Administered Using Either a Weight-based or Weight-banded Approach in Participants With Pulmonary Arterial Hypertension (PAH) on Standard of Care

12. Clinical Framework (PAH Specific)

Primary Condition: Pulmonary Arterial Hypertension (PAH) **Diagnosis Threshold**: WHO Group 1 PAH confirmed by Right Heart Catheterization **Medical Methodology**: Targeted therapy (Sotatercept) administered alongside Standard of Care (Background PAH Therapy) **Optimization Target**: Verification of the safety and PK profile of a simplified Weight-Banded Subcutaneous Dosing regimen

13. Study Procedures & Methodology

Management Pathway: Standard PAH clinical care pathway integrated with investigative dose-titration protocols. **Education Protocol (HCP)**: Site training on administering the correct dosing algorithm (Weight-Based vs. Weight-Banded) and step-up criteria.

Education Protocol (Patient): Guidance on monitoring injection sites and identifying potential AEs. **Quality Audit Mechanism:** Active tracking of patient weight, dose delivered, and PK sampling timelines.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 6 to 9 months (24 - 36 weeks)
Onsite Visit Cadence: Every 3 weeks aligning with the SC injection administration **Remote Monitoring Frequency:** Continuous monitoring of AEs between clinic visits. **Checklist Review Interval:** Prior to every up-titration step during the initial dosing phase.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				